

IP Info API

```
PS C:\Users\trest\OneDrive\Desktop> cd .\Projects\
PS C:\Users\trest\OneDrive\Desktop\Projects> mkdir CI-CD_Deploy-api

Directory: C:\Users\trest\OneDrive\Desktop\Projects

Mode                LastWriteTime         Length Name
----                -
d-----        6/13/2026   3:56 PM             CI-CD_Deploy-api

PS C:\Users\trest\OneDrive\Desktop\Projects> cd CI-CD_Deploy-api
PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> git init
Initialized empty Git repository in C:/Users/trest/OneDrive/Desktop/Projects/CI-CD_Deploy-api/.git/
PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> git branch -m main
PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> git remote add origin https://github.com/Trester17/CI-CD_Deploy-api.git
PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> mkdir -p .github/workflows

Directory: C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api\.github

Mode                LastWriteTime         Length Name
----                -
d-----        6/13/2026   3:58 PM             workflows

PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> mkdir tests

Directory: C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api

Mode                LastWriteTime         Length Name
----                -
d-----        6/13/2026   3:58 PM             tests

PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> python -m venv venv
```

Figure 1: Creating main Project directory.

- First, we will create the main project directory and link this to my github repository which I made specially for this project.

```
PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> venv\Scripts\activate
(venv) PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> pip install -r .\requirements.txt
Collecting flask==3.0.3 (from -r .\requirements.txt (line 1))
  Downloading flask-3.0.3-py3-none-any.whl.metadata (3.2 kB)
Collecting gunicorn==22.0.0 (from -r .\requirements.txt (line 2))
  Downloading gunicorn-22.0.0-py3-none-any.whl.metadata (4.4 kB)
Collecting Werkzeug==3.0.0 (from flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading Werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
Collecting Jinja2>=3.1.2 (from flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.1.2 (from flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.1.3 (from flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading click-8.4.1-py3-none-any.whl.metadata (2.6 kB)
Collecting blinker>=1.6.2 (from flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting packaging (from gunicorn==22.0.0->-r .\requirements.txt (line 2))
  Downloading packaging-26.2-py3-none-any.whl.metadata (3.5 kB)
Collecting colorama (from click>=8.1.3->flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting MarkupSafe>=2.0 (from Jinja2>=3.1.2->flask==3.0.3->-r .\requirements.txt (line 1))
  Downloading MarkupSafe-3.0.3-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
Downloading flask-3.0.3-py3-none-any.whl (101 kB)
Downloading gunicorn-22.0.0-py3-none-any.whl (84 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.4.1-py3-none-any.whl (116 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading markupsafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading packaging-26.2-py3-none-any.whl (100 kB)
Installing collected packages: packaging, MarkupSafe, itsdangerous, colorama, blinker, Werkzeug, Jinja2, gunicorn, click, flask
Successfully installed Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.8 blinker-1.9.0 click-8.4.1 colorama-0.4.6 flask-3.0.3 gunicorn-22.0.0 itsdangerous-2.2.0 packaging-26.2
```

Figure 2: Activating venv (virtual environment) for this python project.

- venv stands for "virtual environment." It's a tool built into Python that creates an isolated copy of Python for a single project. We use command "python -m venv venv" to create new folder called venv which contains its own separate copy of python and its own separate place to install packages which is completely isolated from your computer's main Python installation.

```

PS C:\Users\trest\OneDrive\Desktop\Projects\CI-CD_Deploy-api> pip install pytest pytest-cov bandit pip-audit ruff
Collecting pytest
  Downloading pytest-9.1.0-py3-none-any.whl.metadata (7.6 kB)
Collecting pytest-cov
  Downloading pytest_cov-7.1.0-py3-none-any.whl.metadata (32 kB)
Collecting bandit
  Downloading bandit-1.9.4-py3-none-any.whl.metadata (7.1 kB)
Collecting pip-audit
  Downloading pip_audit-2.10.1-py3-none-any.whl.metadata (28 kB)
Collecting ruff
  Downloading ruff-0.15.17-py3-none-win_amd64.whl.metadata (27 kB)
Requirement already satisfied: colorama>=0.4 in .\venv\Lib\site-packages (from pytest) (0.4.6)
Collecting iniconfig>=2.3.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: packaging>=22 in .\venv\Lib\site-packages (from pytest) (26.2)
Collecting pluggy<2,>=1.5 (from pytest)
  Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Downloading pygments-2.20.0-py3-none-any.whl.metadata (2.5 kB)
Collecting coverage>=7.10.6 (from coverage[toml]>=7.10.6->pytest-cov)
  Downloading coverage-7.14.1-cp314-cp314-win_amd64.whl.metadata (8.8 kB)
Collecting PyYAML>=5.3.1 (from bandit)
  Downloading pyyaml-6.0.3-cp314-cp314-win_amd64.whl.metadata (2.4 kB)
Collecting stevedore>=1.20.0 (from bandit)
  Downloading stevedore-5.8.0-py3-none-any.whl.metadata (2.3 kB)
Collecting rich (from bandit)
  Downloading rich-15.0.0-py3-none-any.whl.metadata (18 kB)
Collecting CacheControl>=0.13.0 (from CacheControl[filecache]>=0.13.0->pip-audit)
  Downloading cachecontrol-0.14.4-py3-none-any.whl.metadata (3.1 kB)
Collecting cyclonedx-python-libc12,>=5 (from pip-audit)
  Downloading cyclonedx_python_lib-11.10.0-py3-none-any.whl.metadata (6.9 kB)
Collecting pip-api>=0.0.28 (from pip-audit)
  Downloading pip_api-0.0.34-py3-none-any.whl.metadata (6.6 kB)
Collecting pip-requirements-parser>=32.0.0 (from pip-audit)
  Downloading pip_requirements_parser-32.0.1-py3-none-any.whl.metadata (9.3 kB)
Collecting requests>=2.31.0 (from pip-audit)
  Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting tomli>=2.2.1 (from pip-audit)
  Downloading tomli-2.4.1-cp314-cp314-win_amd64.whl.metadata (10 kB)
Collecting tomli-w>=1.2.0 (from pip-audit)
  Downloading tomli_w-1.2.0-py3-none-any.whl.metadata (5.7 kB)
Collecting platformdirs>=4.2.0 (from pip-audit)
  Downloading platformdirs-4.10.0-py3-none-any.whl.metadata (5.5 kB)
Collecting license-expression<31,>=30 (from cyclonedx-python-libc12,>=5->pip-audit)
  Downloading license_expression-30.4.4-py3-none-any.whl.metadata (11 kB)
Collecting packageurl-python<2,>=0.11 (from cyclonedx-python-libc12,>=5->pip-audit)
  Downloading packageurl_python-0.17.6-py3-none-any.whl.metadata (5.1 kB)
Collecting py-serializable<3.0.0,>=2.1.0 (from cyclonedx-python-libc12,>=5->pip-audit)
  Downloading py_serializable-2.1.0-py3-none-any.whl.metadata (4.3 kB)
Collecting sortedcontainers<3.0.0,>=2.4.0 (from cyclonedx-python-libc12,>=5->pip-audit)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting boolean.py>=4.0 (from license-expression<31,>=30->cyclonedx-python-libc12,>=5->pip-audit)
  Downloading boolean_py-5.0-py3-none-any.whl.metadata (2.3 kB)
Collecting defusedxml<0.8.0,>=0.7.1 (from py-serializable<3.0.0,>=2.1.0->cyclonedx-python-libc12,>=5->pip-audit)
  Downloading defusedxml-0.7.1-py2.py3-none-any.whl.metadata (32 kB)
Collecting espack<2.0.0,>=0.5.2 (from CacheControl[filecache]>=0.13.0->pip-audit)
  Downloading espack-1.2.0-cp314-cp314-win_amd64.whl.metadata (8.4 kB)
Collecting filelock>=3.8.0 (from CacheControl[filecache]>=0.13.0->pip-audit)
  Downloading filelock-3.29.4-py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: pip in .venv\Lib\site-packages (from pip-api>=0.0.28->pip-audit) (26.1.2)
Collecting pyparsing (from pip-requirements-parser>=32.0.0->pip-audit)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Collecting charset-normalizer<4,>=2 (from requests>=2.31.0->pip-audit)
  Downloading charset_normalizer-3.4.7-cp314-cp314-win_amd64.whl.metadata (41 kB)
Collecting idna>=2.5 (from requests>=2.31.0->pip-audit)
  Downloading idna-3.18-py3-none-any.whl.metadata (6.1 kB)
Collecting urllib3<3,>=1.26 (from requests>=2.31.0->pip-audit)
  Downloading urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)
Collecting certifi>=2023.5.7 (from requests>=2.31.0->pip-audit)
  Downloading certifi-2026.5.20-py3-none-any.whl.metadata (2.5 kB)
Collecting markdown-it-py>=2.2.0 (from rich->bandit)
  Downloading markdown_it_py-4.2.0-py3-none-any.whl.metadata (7.4 kB)
Collecting mdurl>=0.1 (from markdown-it-py>=2.2.0->rich->bandit)
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Downloading pytest-9.1.0-py3-none-any.whl (386 kB)

```

Figure 3: Installing five separate Python packages with one command.

- So we use pip install (as observed in figure 3) to install five different python packages:-
 1. Pytest- it is the testing framework which reads all functions with “test-“ and reports passed/failed for it.
 2. Pytest-cov- add-on (plugin) for pytest that adds code coverage reporting.
 3. Bandit- It is static analysis tool for python which reads source code for known risky pattern without running the code.
 4. Pip-audit- it is dependency vulnerability scanner which checks for known vulnerabilities (like one in figure 5 for Flask CVE (CVE-2026-27205).
 5. Ruff- it is linter which checks for code style and caught the unused request import and missing trailing newlines.

```
(venv) PS C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api> python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.10.6, pytest-7.1.0, pluggy-1.0.0 -- C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api
configfile: pyproject.toml
plugins: cov-2.12.1
collected 11 items

tests/test_app.py::test_valid_ip4 PASSED [ 9%]
tests/test_app.py::test_valid_ip6 PASSED [ 18%]
tests/test_app.py::test_invalid_ip_string PASSED [ 27%]
tests/test_app.py::test_invalid_ip_out_of_range PASSED [ 36%]
tests/test_app.py::test_index PASSED [ 45%]
tests/test_app.py::test_health PASSED [ 54%]
tests/test_app.py::test_valid_public_ip PASSED [ 63%]
tests/test_app.py::test_private_ip PASSED [ 72%]
tests/test_app.py::test_loopback_ip PASSED [ 81%]
tests/test_app.py::test_invalid_ip_route PASSED [ 90%]
tests/test_app.py::test_ipw_loopback PASSED [100%]

===== 11 passed in 0.20s =====
(venv) PS C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api>
```

Figure 4: Running pytest on test_app.py to give report for every “test”.

- We run pytest using command “pytest tests/ -v” to run this inside test folder. And as we can observe in figure 4 result of all different “test-” function which is all passed.

```
(venv) PS C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api> pip-audit -r requirements.txt
Found 1 known vulnerability in 1 package
Name Version ID Fix Versions
-----
flask 3.0.3 CVE-2026-27205 3.1.3
https://pypi.org/project/flask/3.0.3#vulnerabilities https://osv.dev/vulnerability/CVE-2026-27205
```

Figure 5: Error message for known vulnerability in requirements.txt file.

- Pip-audit found a real vulnerability i.e. Flask 3.0.3 has a known CVE (CVE-2026-27205) and the fix for this is to upgrade to Flask 3.1.3. Hence, this is exactly what this tool is *supposed* to do — catch outdated dependencies with known security issues. So, fix it by upgrading from flask 3.0.3 -> 3.1.3 and remove this vulnerability as observed in figure 6.

```
(venv) PS C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api> pip install -r requirements.txt --upgrade
Collecting flask==3.1.3 (from -r requirements.txt (line 1))
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: gunicorn==22.0.0 in .\venv\Lib\site-packages (from -r requirements.txt (line 2)) (22.0.0)
Requirement already satisfied: blinker==1.9.0 in .\venv\Lib\site-packages (from flask==3.1.3->-r requirements.txt (line 1)) (1.9.0)
Requirement already satisfied: click>=8.1.3 in .\venv\Lib\site-packages (from flask==3.1.3->-r requirements.txt (line 1)) (8.4.1)
Requirement already satisfied: itsdangerous>=2.2.0 in .\venv\Lib\site-packages (from flask==3.1.3->-r requirements.txt (line 1)) (2.2.0)
Requirement already satisfied: Jinja2>=3.1.2 in .\venv\Lib\site-packages (from flask==3.1.3->-r requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: MarkupSafe>=2.1.1 in .\venv\Lib\site-packages (from flask==3.1.3->-r requirements.txt (line 1)) (3.0.3)
Requirement already satisfied: Werkzeug>=3.1.0 in .\venv\Lib\site-packages (from flask==3.1.3->-r requirements.txt (line 1)) (3.1.8)
Requirement already satisfied: packaging in .\venv\Lib\site-packages (from gunicorn==22.0.0->-r requirements.txt (line 2)) (26.2)
Requirement already satisfied: colorama in .\venv\Lib\site-packages (from click>=8.1.3->flask==3.1.3->-r requirements.txt (line 1)) (0.4.6)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Installing collected packages: flask
  Attempting uninstall: flask
    Found existing installation: Flask 3.0.3
    Uninstalling Flask-3.0.3:
      Successfully uninstalled Flask-3.0.3
  Successfully installed flask-3.1.3
(venv) PS C:\Users\trent\OneDrive\Desktop\Projects\CI-CD_Deploy-api> pip-audit -r requirements.txt
No known vulnerabilities found
```

Figure 6: Upgrading flask 3.0.3 to 3.1.3 will fix CVE-2026-27205 vulnerability.

```

(venv) PS C:\Users\trext\OneDrive\Desktop\Projects\CI-CD_Deploy-api> bandit -r app.py -ll
[main] INFO profile include tests: None
[main] INFO profile exclude tests: None
[main] INFO cli include tests: None
[main] INFO cli exclude tests: None
[main] INFO running on Python 3.14.6
Run started:2026-06-13 22:19:22.931217+00:00

Test results:
>> Issue: [B104:hardcoded_bind_all_interfaces] Possible binding to all interfaces.
Severity: Medium Confidence: Medium
CWE: CWE-605 (https://cwe.mitre.org/data/definitions/605.html)
More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b104\_hardcoded\_bind\_all\_interfaces.html
Location: .\app.py:44:17
43     debug = os.environ.get("FLASK_DEBUG", "false").lower() == "true"
44     app.run(host="0.0.0.0", port=port, debug=debug)

-----

Code scanned:
Total lines of code: 32
Total lines skipped (#nosec): 0

Run metrics:
Total issues (by severity):
    Undefined: 0
    Low: 0
    Medium: 1
    High: 0
Total issues (by confidence):
    Undefined: 0
    Low: 0
    Medium: 1
    High: 0

Files skipped (0):
(venv) PS C:\Users\trext\OneDrive\Desktop\Projects\CI-CD_Deploy-api> ruff check .
>>
F401 [*] `flask.request` imported but unused
--> app.py:1:35
|
1 | from flask import Flask, jsonify, request
|                                     ^^^^^^^
2 | import ipaddress
3 | import os
|
help: Remove unused import: `flask.request`

W292 [*] No newline at end of file
--> app.py:44:52
|
42 |     port = int(os.environ.get("PORT", 5000))
43 |     debug = os.environ.get("FLASK_DEBUG", "false").lower() == "true"
44 |     app.run(host="0.0.0.0", port=port, debug=debug)
|                                     ^
|
help: Add trailing newline

W292 [*] No newline at end of file
--> tests\test_app.py:75:39
|
73 |     data = resp.get_json()
74 |     assert data["version"] == "IPv6"
75 |     assert data["is_loopback"] is True
|                                     ^
|
help: Add trailing newline

Found 3 errors.
[*] 3 fixable with the `--fix` option.

```

Figure 7: Errors from both tests bandit and ruff.

- We ran two different tests- bandit and ruff both of which gives error. So, to fix it we can implement:-
1. Bandit found 1 issue (Medium severity)- B104: Possible binding to all interfaces which is about host="0.0.0.0" in app.py line 44. Bandit flags this because binding to 0.0.0.0 means the app accepts connections from any network interface, not just localhost which is correct for a containerized app, but bandit flags it as worth a second look. It is actually fine to keep it but since bandit flags it, we should acknowledge it explicitly using a # nosec comment using command "app.run(host="0.0.0.0", port=port, debug=debug) # nosec B104 - required for containerized deployment"
 2. Ruff found 3 issues which all are trivial and all auto-fixable. So, ruff can fix it automatically using "ruff check . -fix".

```
PS C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api> ruff check .
Found 3 errors (3 fixed, 0 remaining)
(cero) PS C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api> bandit -r app.py -ll
[main] INFO profile include tests: None
[main] INFO profile exclude tests: None
[main] INFO cli include tests: None
[main] INFO cli exclude tests: None
[main] INFO running on Python 3.14.0
[manager] WARNING Test in comment: required is not a test name or id, ignoring
[manager] WARNING Test in comment: for is not a test name or id, ignoring
[manager] WARNING Test in comment: containerized is not a test name or id, ignoring
[manager] WARNING Test in comment: deployment is not a test name or id, ignoring
Run started:2025-06-12 22:25:27.066633+00:00

Test results:
No issues identified.

Code scanned:
Total lines of code: 32
Total lines skipped (#nosec): 0

Run metrics:
Total issues (by severity):
  Undefined: 0
  Low: 0
  Medium: 0
  High: 0
Total issues (by confidence):
  Undefined: 0
  Low: 0
  Medium: 0
  High: 0

Files skipped (0):
(cero) PS C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api> ruff check .
All checks passed!
(cero) PS C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api> python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.14.6, pytest-9.1.0, pluggy-1.6.0 -- C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api
configfile: pyproject.toml
plugins: cov-7.1.4
collected 11 items

tests/test_app.py::test_valid_ip_v4 PASSED [ 9%]
tests/test_app.py::test_valid_ip_v6 PASSED [ 18%]
tests/test_app.py::test_invalid_ip_string PASSED [ 27%]
tests/test_app.py::test_invalid_ip_out_of_range PASSED [ 36%]
tests/test_app.py::test_invalid_ip PASSED [ 45%]
tests/test_app.py::test_health PASSED [ 54%]
tests/test_app.py::test_valid_public_ip PASSED [ 63%]
tests/test_app.py::test_private_ip PASSED [ 72%]
tests/test_app.py::test_loopback_ip PASSED [ 81%]
tests/test_app.py::test_invalid_ip_route PASSED [ 90%]
tests/test_app.py::test_ip_v6_loopback PASSED [100%]

===== 11 passed in 0.13s =====
(cero) PS C:\Users\trout\OneDrive\Desktop\Projects\CI-CD_Deploy-api>
```

Figure 8: Perfect 3 different checks without any errors.

- So, now in figure 8 we can see that finally there are no errors in all three different tests (pytest, bandit and ruff) after we fixed all errors and vulnerabilities.

Require all commits be made to a non-target branch and submitted via a pull request before they can be merged.

☒ **Require status checks to pass**
Choose which status checks must pass before the ref is updated. When enabled, commits must first be pushed to another ref where the checks pass.

Hide additional settings ^

☐ **Require branches to be up to date before merging**
Whether pull requests targeting a matching branch must be tested with the latest code. This setting will not take effect unless at least one status check is enabled.

☐ **Do not require status checks on creation**
Allow repositories and branches to be created if a check would otherwise prohibit it.

Status checks that are required + Add checks ▾

Run Tests	GitHub Actions	🗑
Lint & Format Check	GitHub Actions	🗑
Dependency Vulnerability Scan	GitHub Actions	🗑
Static Analysis (Bandit)	GitHub Actions	🗑
Detect Leaked Secrets (GitLeaks)	GitHub Actions	🗑

☒ **Block force pushes**
Prevent users with push access from force pushing to refs.

☐ **Require code scanning results**
Choose which tools must provide code scanning results before the reference is updated. When configured, code scanning must be enabled and have results for both the commit and the reference being updated.

☐ **Require code quality results**
Choose which severity levels of code quality results should block pull request merges. When configured, a code quality analysis must be done on the pull request before the changes can be merged.

☐ **Automatically request Copilot code review**
Request Copilot code review for new pull requests automatically if the author has access to Copilot code review and their premium requests quota has not reached the limit.

Create

Figure 9: Enable “Require status checks to pass” option for better main-branch-repository security.

- We add two additional securities for main-branch-repository (as mentioned in README.md) :-
 1. Add dependabot which is Github’s build-in bot which once configured, it automatically checks your dependencies on a schedule and opens a pull request

whenever newer version of package exists and package has known security vulnerability and fixed version is available.

2. Add branch protection on main branch repository. This setting restricts what can happen to main branch by adding specific rules like: (as observed in figure 9)
 - a) Run Tests
 - b) Lint & Format Check
 - c) Dependency Vulnerability Scan
 - d) Static Analysis (Bandit)
 - e) Detect Leaked Secrets (Gitleaks).